DDPM Fusing Mamba and Adaptive Attention: A Augmentation Method for Industrial Control Systems Anomaly Data

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Abstract

The security monitoring of Industrial Control Systems heavily relies on accurate anomaly detection techniques. However, the field faces a key challenge: the extreme scarcity of anomalous samples available for training, primarily due to issues of data privacy protection and system security. Generating synthetic data using generative models is one of the effective ways to alleviate the scarcity of abnormal data. However, when dealing with the complex characteristics of Industrial Control Systems data, such as strong periodicity, the interdependence, and correlation among multiple variables, existing generative models often fail to accurately capture long temporal dependencies or preserve localized detail features. This frequently results in synthetic data that lack overall consistency with the true underlying distribution. To address these issues, this study proposes ETUA-DDPM, an anomaly data augmentation method based on the Denoising Diffusion Probabilistic Model. This method leverages the advantages of diffusion models, namely stable training and consistency in the generated distribution, to achieve high-fidelity synthesis of scarce abnormal data. Crucially, the state-space model Mamba is integrated into the DDPM's denoising network to accurately model and capture the data's long temporal dependencies, thereby

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ensuring a precise match with the global distribution. Furthermore, a multi-scale local detail feature and an adaptive attention adjustment mechanism are introduced during the information flow to compensate for critical abnormal details that may be lost during the feature aggregation process. Evaluation based on similarity metrics and downstream anomaly detection performance demonstrates that ETUA-DDPM can generate anomalous industrial control system time series data with superior distributional fidelity and more granular details, thus effectively alleviating the sample scarcity problem and markedly boosting performance in real-world applications.

Keywords: industrial control systems, anomaly detection, multivariate time series data, generation, diffusion model, mamba, precise distribution matching

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1. Introduction

Industrial Control Systems (ICS) are the core operational support for critical infrastructures such as electricity, water, and transportation. With the deep integration of the industrial Internet and information technology, the openness and interconnectivity of ICS have continuously increased, but they also face increasingly severe cybersecurity threats [1]. Notorious cyberattacks, including Stuxnet, Night Dragon, Duqu, Flame, Black Energy, NotPetya, Dragonfly, and the Gas Pipeline Cyber-Intrusion, have all targeted ICS [2]. Anomaly detection is a critical technology for ensuring the secure operation of these systems, and its effectiveness directly determines the capability to identify and respond to potential attacks. Currently, most advanced anomaly detection models (especially deep learning models) are data-driven, meaning their performance is highly dependent on large and representative training datasets.

In practical industrial scenarios, factors such as system security requirements, privacy protection constraints, and the inherently low frequency of attack incidents result in an extreme scarcity of anomalous samples available for training. This severely limits the generalizability and robustness of detection models. To mitigate this issue, data generation methods based on generative models have emerged as a crucial technical pathway to expand anomalous samples and improve model performance. However, ICS time-series data, particularly Industrial Multivariate Anomalous Time-series Data (IMATD), exhibit strong periodicity, high regularity, and complex interdependencies and correlations among variables. Consequently, generative models face a dual challenge: on the one hand, they must faithfully restore the overall distribution and long temporal dependencies of the data at a macroscopic level; on the other hand, they need to precisely retain abnormal details and the interdependence and correlation among variables at a microscopic level. Existing generative methods, such as Variational Autoencoders (VAE) [3], learn the latent space distribution of the data for reconstruction. However, when the data distribution is complex, VAEs tend to generate smooth, averaged samples to cover various possibilities, leading to blurring and distortion of detail. Furthermore, they struggle to fully capture the complex coupling relationships between multiple variables, resulting in insufficient intervariate correlation.Generative Adversarial Networks (GANs) [4] learn the data distribution through the adversarial game between a generator and a discriminator. However, the loss functions of the generator and discriminator often oscillate violently, making it difficult to find a stable equilibrium point. This frequently leads to issues such as vanishing gradients, resulting in training instability. Alternatively, the generator may discover that certain outputs can effectively deceive the current discriminator, causing it to generate repeatedly identical or highly similar samples. This phenomenon, known as mode collapse, results in generated data deviating from the true distribution [5, 6].

Denoising Diffusion Probabilistic Models (DDPMs) [7] are Markov chain-based generative models that approximate the original real data distribution by learning the reverse diffusion process. They possess inherent advantages such as stable training and strong distribution fit capabilities, which theoretically ensure the consistency of the generated data with the original distribution. However, due to the characteristics of IMATD, including strong periodicity and multivariate coupling, traditional DDPMs, whose backbone, the U-Net, is restricted by the limited, local receptive field of its convolutional operations, struggle to accurately model the long temporal dependencies within IMATD. Furthermore, the multi-layer downsampling and information aggregation processes inevitably lead to the compression and loss of detailed features. To address these challenges, this paper proposes ETUA-DDPM, a DDPM-based anomaly data augmentation method designed to achieve IMATD generation that unifies distributional accuracy with detail preservation, thereby enabling effective augmentation of industrial control multivariate anomaly data. Specifically, we integrate the state space model Mamba [8] into the U-Net architecture of DDPM. By leveraging Mamba's characteristics of high efficiency, linearity, and global receptive field, we can accurately capture the IMATD's strong periodic patterns and long temporal dependencies, thus improving the overall distribution matching accuracy of the generated IMATD. Simultaneously, we introduce a multi-scale detail capture and an attention weighting mechanism to dynamically adjust the contribution of the encoder information flow. This selectively compensates for and preserves fine-grained features and intervariable correlations in the abnormal data, ultimately strengthening the microscopic realism of the generated samples. The main contributions of this paper can be summarized as follows:

1. The ETUA-DDPM framework is proposed, which leverages the inherent distribution-fitting strengths of DDPM as its foundation to generate industrial multivariate anomalous time series data that are highly consistent with the true distribution. This effectively augments the abnormal sample set, thereby enhancing the capabilities of ICS anomaly detection models.
2. An Enhanced Long Temporal Dependency Capture U-Net (ETU) is proposed. By integrating residual convolution with the Mamba module, the network maintains local feature extraction capability while strengthening the global modeling of strong periodicity and long temporal dependencies. This ensures a precise matching of the overall distribution of the synthesized data.
3. An Adaptive Attention Skip Connection (AASC) is designed. This mecha-

nism employs multiscale local features and a dynamic attention weighting scheme to assign weights to features at each timestep, dynamically adjusting the information flow. This significantly improves the model's ability to preserve and reconstruct fine-grained details, further elevating the quality of the generated samples.

4. Comprehensive experiments involving similarity metrics and downstream anomaly detection tasks validate that the proposed method significantly expands scarce anomaly datasets and significantly improves the performance of the detection model.

The remainder of this paper is organized as follows. Section 2 discusses existing research on the generation of time series or abnormal time series data within the ICS environment and addresses their limitations. Section 3 introduces the proposed anomaly data generation method. Section 4 presents the experimental results, including comparisons of distribution similarity between the original and generated anomaly data sets, along with validation of downstream anomaly detection. Section 5 concludes the paper and describes future research directions.

2. Related Works

2.1. Industrial Control Time-series Data Generation Method

The scarcity of anomalous data in ICS environments has spurred the development of various generation techniques. For instance, some studies have focused on non-deep learning approaches for data generation. Lee et al. proposed an ICS anomaly data generation method based on systematic sampling and linear regression models [9], which improves global correlation and physical plausibility in the generated data. However, this method fundamentally relies on feature value adjustment rather than deep temporal pattern learning, making it difficult to capture complex non-linear temporal dependencies. Kim et al. developed a data-driven approach for the generation of ICS anomaly data that reconstructs the entire ICS environment within virtualized containers to produce synthetic data [10]. However, this method heavily depends on prior knowledge of the ICS system topology and control logic.

Researchers have also explored the use of generative models to achieve time-series data generation. Klopries et al. proposed a method for synthesizing time series data to be used in unsupervised autoencoder training [11], aiming to mitigate real data scarcity by generating high-quality synthetic data. Li et al. introduced the Causal Recurrent Variational Autoencoder (CR-VAE) [12], which incorporates underlying causal mechanisms by learning Granger causality in multivariate time series for data generation. Li et al. proposed DT-VAE [13], which combines generalizable domain knowledge with a cumulative difference learning mechanism to effectively reduce the accumulation of errors during the generation process. Jeon et al. presented a multivariate time series anomaly generation method based on latent-space perturbation and a recurrent variational autoencoder based on the attention mechanism [14]. This method effectively captures temporal dependencies and multivariate correlations in ICS, generating visually and statistically credible abnormal data, thus mitigating the class imbalance problem and improving the detection performance of binary classifiers. Although improvements in VAE-based models have enhanced their stochastic sampling capability, they still face issues such as blurred, noisy, and poorly similar generated data, which limits their application in industrial control data generation. Yang et al. proposed TS-GAN [15], which integrates GAN with a Long Short-Term Memory (LSTM) [16] network to improve the understanding of spatial dependencies, generating high-fidelity time series data with the same temporal and spatial dependencies as the real data. Xu et al. introduced TGAN-AD [17], which overcomes the limitations of traditional GANs in capturing long-range contextual dependencies and parallel computing capabilities for time series by incorporating a Transformer. Klopries et al. proposed ITF-GAN [18], which addresses the deficiencies of traditional GANs in feature manipulation and interpretability through explainable feature generation and control. Furthermore, Han et al. introduced SGAN [19], a semi-supervised GAN that utilizes limited labeled data in ICS to generate synthetic data highly similar to the real data distribution. Yuan et al. proposed B-GAN [20], a data balancing method based on a Generative Adversarial Network and LSTM, to enhance the training set by generating synthetic abnormal samples. However, the inherent drawbacks of GANs include training instability, and their adversarial learning mechanism struggles to ensure that the global data distribution of the generated samples perfectly aligns with the scarce real abnormal data. Consequently, abnormal data generated by GANs often achieve only local morphological similarity, but deviate from seed samples in terms of overall behavioral distribution [6].

Denoising Diffusion Probabilistic Models (DDPMs) are a deep learning method that has been widely applied in the field of generative modeling in recent years, achieving excellent results [21]. Their core advantages lie in high training stability and strong distribution fitability. DDPM simulates a gradual diffusion process via a Markov chain, progressively denoising the data from a simple distribution (such as a standard Gaussian distribution) to the complex real data distribution. This process enables the model to approximating the data in a stably and unbiased way and to learn the true global distribution of the data. For example, Ren et al. proposed Diff-MTS [22], a conditional adaptive diffusion model with enhanced temporal dependency modeling, designed to better handle the temporal dynamics characteristic of industrial control time series data. However, when dealing with multivariate time series data (exceeding 50 variables), it still faces challenges in precisely matching the overall distribution.

2.2. Temporal Dependency Modeling Method

State Space Sequence Models (SSMs) [23] are a class of frameworks specifically designed for modeling sequence data. By mapping the input to a hidden state before generating the output, they can effectively capture the dynamic patterns and long temporal dependencies inherent in sequences. Structured State-Space Sequence Models (S4) [24] address the issues of numerical instability and excessive computational complexity in long temporal dependency modeling within SSMs by imposing specific structural constraints on the state matrix.

This significantly improves model performance in long-sequence tasks involving ICS data while preserving theoretical advantages. Building upon this, Mamba introduces a selective state-space mechanism, allowing its parameters to change adaptively based on the input content. This overcomes the limitations of linear models like S4 in terms of content awareness, allowing sensitive perception and adaptive modeling of contextual information. Furthermore, Mamba, through a simplified hardware-aware algorithm and the integration of components such as one-dimensional causal convolution and normalization layers, efficiently aggregates global information and captures the long temporal dependencies of IMATD while inheriting the long-sequence modeling capabilities of S4.

3. DDPM FUSING MAMBA AND ADAPTIVE ATTENTION: ETUA-DDPM

3.1. Denoising Diffusion Probabilistic Model

DDPM is a generative model based on Markov chains, inspired by thermodynamic processes. As shown in Fig. 1, its fundamental idea is to gradually add noise, transforming the original data into pure noise, thus learning the latent distribution of the target data. This process involves two stages: forward diffusion and reverse denoising.

In the forward diffusion process, Gaussian noise is progressively added to the original data $\mathbf{X}_0$ over T steps, resulting in $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_T$, until the data are completely transformed into Gaussian noise at the termination step T. Each noise addition step can be viewed as a Markov chain:

$$q(X_t|X_{t-1}) = \mathcal{N}(X_t; \sqrt{1-\beta_t}X_{t-1}, \beta_t\mathbf{I}) \tag{1}$$

Here, β_t represents the table of noise coefficients. Through reparameterization and the additive property of the normal distribution, the addition of noise can be expressed as:

$$X_t = \sqrt{1-\alpha_t}\varepsilon_t + \sqrt{\alpha_t}X_{t-1} = \sqrt{1-\bar{\alpha}_t}\bar{\varepsilon}_t + \sqrt{\bar{\alpha}_t}X_0 \tag{2}$$

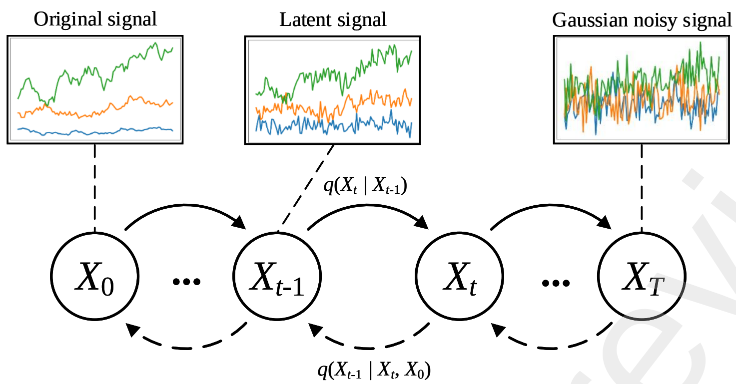

Figure 1: Flowchart of the ETUA-DDPM Time series Data Generation Process.

Here, ε_t represents random noise that follows a standard normal distribution, $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$.

In the reverse denoising process, the original data $\mathbf{X}_0$ progressively restored from the pure noise sample $\mathbf{X}_t$ through a step-by-step denoising procedure. This process relies on probabilistic predictions of the reverse distribution in order to accurately recover the data structure by optimizing the noise removal process. Assuming the current state $\mathbf{X}_0$ is known, the formula for the reverse denoising process can be derived based on Bayes' Theorem as follows:

$$q(X_{t-1}|X_t, X_0) = \mathcal{N}(X_{t-1}; \bar{\mu}(X_t, X_0), \bar{\beta}_t \mathbf{I}) \tag{3}$$

where the mean value is:

$$\bar{\mu}(X_t, X_0) = \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \varepsilon_t \right) \tag{4}$$

By progressively optimizing the reverse process, DDPM is able to effectively generate data samples.

3.2. Mamba

S4 is a widely used mathematical framework for sequence modeling whose core idea is to map input data to the output signal by introducing a latent state vector. This mapping relationship is typically expressed through a system of linear ordinary differential equations (ODEs):

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{aligned} \tag{5}$$

The state matrices $A \in \mathbb{R}^{N \times N}$, B, and $C \in \mathbb{R}^{N}$ are the parameters, while $h'(t)$ represents the derivative of $h(t)$ with respect to time t. The Mamba module is a deep learning model built upon S4, which utilizes the zero-order hold (ZOH) rule for discretization to iteratively solve the above formula:

$$\begin{aligned} &\bar{A} = \exp(\Delta A); \quad \bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \times \Delta B \\ &h_t = \bar{A}h_{t-1} + \bar{B}x_t; \quad y_t = Ch_t \end{aligned} \tag{6}$$

By dynamically parameterizing key parameters of the SSM through the input temporal data, the model enables an input-dependent selective mechanism, thereby enhancing its ability to model long temporal dependencies.

3.3. ETUA-DDPM Temporal Data Augmentation Method

To generate IMATD highly consistent with the true distribution, this paper proposes an industrial control multivariate abnormal time-series data augmentation method-ETUA-DDPM. This method primarily enhances the diffusion model's capability for long temporal relationship modeling by implementing an Enhanced Long Temporal Dependency Modeling U-Net (ETU), thereby ensuring the diffusion model can learn the global distribution of IMATD. Simultaneously, we design an Adaptive Attention Skip Connection (AASC) to guarantee the precise capture of detailed features during the feature fusion process, mitigating the loss of details during temporal dependency modeling.

The overall process of ETUA-DDPM time series data generation is illustrated in Fig. 2. First, as shown in Forward Processing, the raw IMTSD data undergoes preprocessing through normalization. Then, noise is added to the data through the forward diffusion process, generating a noisy data set suitable for model training. Next, the processed data set is fed into the ETU network, where long temporal dependency modeling and detail extraction are performed

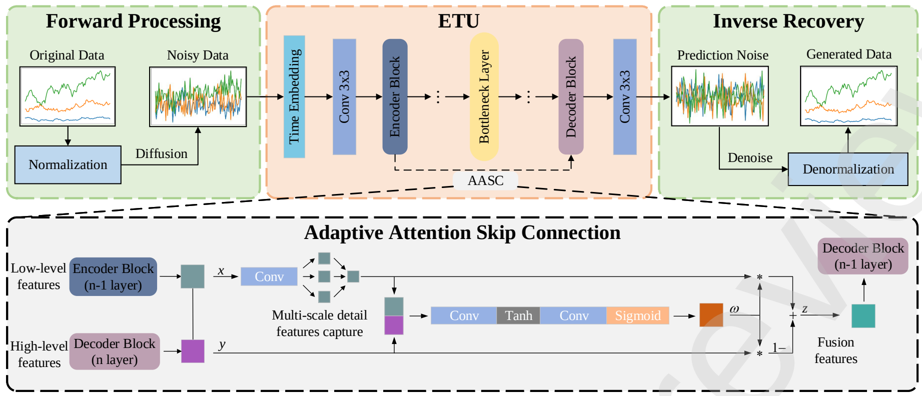

Figure 2: Flowchart of the ETUA-DDPM Time series Data Generation Process.

in conjunction with AASC, leading to preliminary prediction results. Finally, in Inverse Recovery, these prediction results are used for reverse denoising and recovery, and inverse normalization is applied to generate IMATD.

3.4. Introduction to ETUA Structure

Current U-Net-based diffusion models primarily rely on local convolutional kernels to extract local features, focusing on information extraction within local neighborhoods. However, the limited receptive field of these convolutional kernels restricts their ability to model long temporal dependencies. Although the skip connections in U-Net effectively integrate low-level details with high-level features, these connections mainly focus on the fusion of local information and do not sufficiently enhance the network's ability to model global context and long temporal dependencies. This limitation is particularly pronounced in long-sequence modeling tasks.

To address this issue, this paper proposes an ETU architecture (as shown in Fig. 3). First, a network architecture with a residual learning mechanism is introduced (as shown in part a of Fig. 3), which effectively captures the temporal variations in sequential data through convolution operations and residual learning, thereby accurately extracting local patterns. Next, the selective state-space mechanism of Mamba is integrated (as shown in part b of Fig. 3) to provide

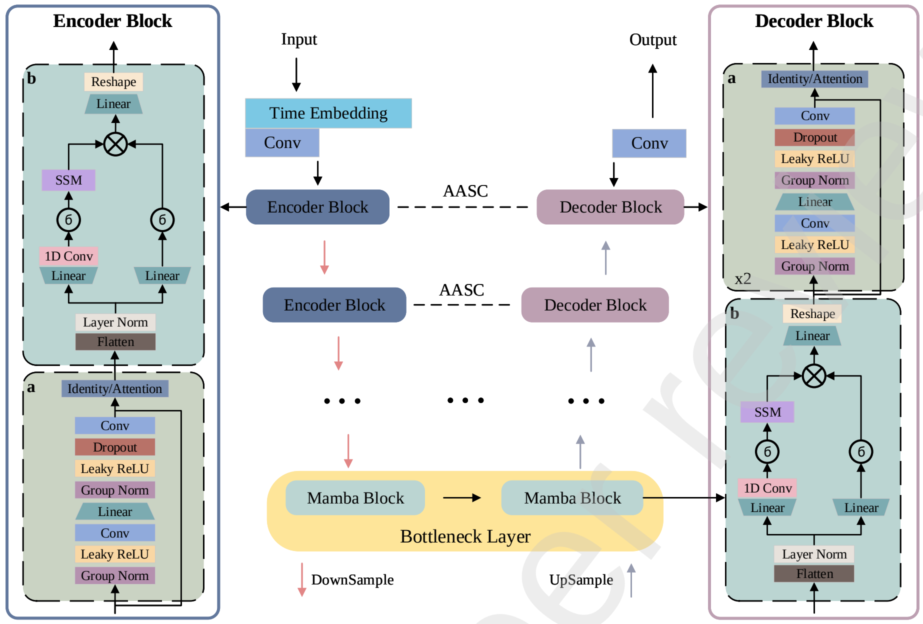

Figure 3: Flowchart of the ETUA-DDPM Time series Data Generation Process.

the network with a global receptive field and aggregate contextual information throughout the input sequence. This enables the model to learn deeper long-term temporal dependencies, thereby enhancing its capability to model these dependencies. Through the design of modules a and b, the encoder Block and the decoder Block modules are developed, and by combining these, the encoder, bottleneck layer, and decoder are reconstructed, ultimately forming the ETU network.

It is important to note that, to capture long temporal dependencies, features over a long time range are aggregated during the downsampling and upsampling processes, which leads to the loss of some details. To address this issue, we designed the AASC (as shown in Fig. 3). The AASC first applies convolution to match channels in the lower level features, allowing the model to flexibly adjust the information channel structure at different levels, thus improving the network adaptability. Furthermore, the convolution operation introduces multiscale local detail information from the lower-level features, ensuring that the network can capture both richer local details and higher-level semantic information during the feature fusion process. Building on this, we introduce an attention mechanism [25], which weights the features at each time step, adaptively adjusting the information flow according to the importance of the features at different levels. This ensures that the feature fusion process is both more efficient and accurate, ultimately achieving the precise capture of detailed features, thus minimizing information loss and enhancing the model's capability for detail extraction.

3.5. Implementation Method of ETU

3.5.1. The encoder section

We designed an encoder block module that is tailored to the characteristics of the IMTSD data. First, we apply GroupNorm and Leaky ReLU activation functions to normalize and apply nonlinear activation to the input, thereby stabilizing the training process and enhancing the model's expressive power. Next, convolutional layers are used to extract local features from the input. Subsequently, a fully connected layer is employed for temporal embedding, injecting temporal information into the model to help capture long-range temporal dependencies in the sequential data. Then, GroupNorm and Leaky ReLU are applied again to normalize and activate the features embedded with temporal information, further accelerating the model's training and improving its stability. After that, convolution operations are used to extract higher-level temporal features, and a dropout layer is introduced to reduce overfitting and enhance the model's generalization ability. Finally, residual connections and an attention mechanism are employed to further improve the feature expression ability, helping the model capture long temporal relationships.

After the extracted feature information is flattened and transposed, it is fed into the b module, which consists of two parallel branches. In the first branch, the features are expanded through a linear layer, followed by convolution via a 1D convolutional layer, with the SiLU activation function applied. The features are then modeled using the SSM layer to capture the correlation between local features. In the second branch, the features are also expanded, processed through a linear layer and SiLU activation function, and used to model long temporal dependencies. The outputs from the two branches are fused via the Hadamard product, effectively integrating the two different levels of temporal features. Finally, the fused features are reshaped and transposed through a linear layer, restoring them to their original shape. The purpose of this sequence of operations is to effectively enhance the complementarity between the features from the two branches, thereby better capturing long temporal dependencies in the time series data.

3.5.2. The bottleneck layer

We adopt two consecutive blocks of b for deep processing, aiming to further compress the high-dimensional features and extract the higher-level features. This enables the features input into the decoder to be more enriched, allowing the model to fully explore complex long temporal dependencies at the bottleneck stage.

3.5.3. The decoder part

We follow the symmetric structure of U-Net and design the Decoder Block module. During the reconstruction of the time series data, the b module first helps the model capture complex temporal dependencies in IMTSD data from historical information, thereby enabling more accurate modeling of long temporal dependencies. This strengthens the expressive power of the decoder, allowing it to better recover and generate the target time-series data. Then, two residual blocks are applied. The goal is to further refine the high-level abstract features generated by the b module, improving the recovery of local details, and reducing potential error accumulation. The first residual block helps the model extract finer local features by further learning the relationships between features and supplementing potential detail information. The second residual block further refines the extracted features, helping the network recover more specific and detailed long temporal dependencies from the more abstract features.

3.6. Implementation Method of AASC

We designed the Adaptive Attention Skip Connection (AASC), as shown in Fig. 2. First, the low-level features $\mathbf{x}$ from the encoder are convolved, and the channels of $\mathbf{x}$ match the high-level features $\mathbf{y}$. Simultaneously introducing multiscale local detail information of $\mathbf{x}$ and processing it with high-level features $\mathbf{y}$ of the dexcoder. This process learns an adaptive feature fusion weight $\mathbf{w}$ that adjusts the fusion ratio between $\mathbf{x}$ and $\mathbf{y}$, effectively transmitting key information and compensating for the potential loss of detail during modeling temporal dependencies. The computation is as follows:

$$z = (1 - w)x + wy \tag{7}$$

Here, $w \in (0, 1)$ represents the weight value and z denotes the newly fused feature.

This approach improves the precision of the capture details, thereby significantly boosting the performance of the diffusion model.

3.7. Algorithm Description

The data generation method based on ETUA-DDPM is divided mainly into two parts: training the model and generating data. The entire training process is described in Algorithm 1.

First, the model parameters are initialized and the original data set $\mathbf{X}$ is preprocessed to obtain X'. The processed data are then fed into the model to predict noise, and the error between the true and predicted noise is computed by Eq. (10). Finally, the AdamW optimization algorithm is employed as the optimizer to update the model by backpropagation, based on the calculated error.

The algorithm for the data generation process is outlined in the Algorithm Table 2. Initially, samples x_T are randomly generated from a Gaussian noise distribution and fed into the trained model. Using Eq. (3)-(4) to obtain denoised data. Perform noise prediction first and obtain denoised data $\bar{\mu}(x_t, x_0)$ based on the predicted noise $\varepsilon_\theta(x_t, t)$. This process is repeated for a total of

Algorithm 1 : Training the ETUA-DDPM

Initial parameter: Industrial control dataset X, diffusion total time step T, number of training rounds *epochs*, batch size B, noise schedules $\{\beta_t\}_{t=1}^{T}$.

Preprocess $x_i \sim X$:

$$X' \leftarrow 2 \times (x_i - \min(x))/(\max(x) - \min(x)) - 1$$

for i in range(*epochs*) **do**

  for j in range(B) **do**

    Sampling initial data $x_0 \sim X'$, Gaussian noise $\varepsilon \sim \mathcal{N}(0, I)$ and diffusion time step $t \sim \text{Uniform}(1, T)$

    Calculate x_t after adding noise:

    $x_t \leftarrow \sqrt{1 - \bar{\alpha}_t}\varepsilon_t + \sqrt{\bar{\alpha}_t}x_0$

    Predict the increased noise $\varepsilon_\theta(x_t, t)$ during the diffusion process;

    Compute loss of the model:

    $loss \leftarrow MSE(\varepsilon, \varepsilon_\theta(x_t, t))$

  end for

  Update model parameters by minimizing (AdamW Optimizer) *loss*;

end for

T diffusion time steps to generate the samples x_0 for further processing. Finally, the industrial control time series dataset X'' is obtained through a data restoration operation.

4. EXPERIMENTATION AND EVALUATION

In this section, we provide a detailed description of the datasets used in the experiments and the corresponding evaluation methods. To validate the effectiveness of the proposed method, we compare the ETDU-DDPM method with several other mainstream generative models, including TimeGAN [26], TimeVAE [27], Diff-MTS, DiffWave and tabddpm [28]. Specifically, we analyze the similarity between the data generated by these models and the original data in terms of characteristics such as distribution and diversity. Furthermore, we assess the effectiveness of the ETUA-DDPM method by evaluating the prac-

Algorithm 2 Generate Time series Data

1: **for** i in range($epochs$) **do**
2: Sample x_T from $\mathcal{N}(0, I)$;
3: **for** t in range($T, 0, -1$) **do**
4: Predict the increased noise $\varepsilon_\theta(x_t, t)$;
5: Calculate the denoised data by predicting noise:
6: $\bar{\mu}(x_t, x_0) \leftarrow \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t, t)\right)$,
7: $x_{t-1} \leftarrow \mathcal{N}(x_{t-1}; \bar{\mu}(x_t, x_0), \bar{\beta}_t I)$;
8: **end for**
9: Generate time series data x_0
10: **end for**
11: Perform data restoration operation on x_0 to obtain the final generated time series dataset X''

tical performance (i.e., accuracy, F1 score, and area under the ROC curve) of several widely used machine learning algorithms on the data generated by each model.

Through comparative analysis, we explore the performance of each model in terms of data generation quality, providing an in-depth evaluation of the differences and advantages in the generated results, with a particular focus on the applicability of ETUA-DDPM in IMTSD generation. In addition, machine learning-based evaluation methods further confirm the effectiveness of the proposed method in practical applications, offering both theoretical justification and empirical support for its use in real-world scenarios.

4.1. Experimental Dataset

The experiment uses the publicly available SWaT [29] and WADI [30] for evaluation and testing.

The SWaT (Secure Water Treatment) dataset is a publicly available industrial control system security research dataset designed to simulate a real-world water treatment system. This data set has 51 columns of eigenvalues divided into two categories: one containing anomalous data with various types of attacks and the other containing data from normal operations. It is commonly used to evaluate the effectiveness of security solutions such as intrusion detection systems and security event response systems.

The WADI (Water Distribution Network Dataset) is another publicly available industrial control system security research dataset, simulating a real-world water distribution system. It has 123 columns of eigenvalues, containing a large amount of sensor data and network traffic data, which can be used to analyze abnormal behavior and attacks within the water distribution system. This data set is also divided into two categories: anomalous and normal data.

Anomalous data from these two publicly available datasets are used as seed samples to generate time-series data, which are then combined with a small amount of original data to create a new dataset for training machine learning models.

4.2. Evaluation Indicators

We input the training sets from two different datasets into the generative model for training and use the model weights obtained after training to generate data. The evaluation of the generated data is conducted primarily through a comprehensive analysis based on the following two dimensions:

1. Evaluation of the similarity between the generated data and the original data: The generated data are evaluated from four aspects: distribution similarity, morphological similarity, feature correlation, and diversity. Let G and R represent the generated sample matrix and the actual sample matrix, respectively, and let g_i and r_i represent the i-th row of the generated sample matrix and the i-th row of the original sample matrix, respectively. Based on this, the following eq. (8)–(13) are used for quantitative measurement, assessing the closeness of the data distribution, the structural similarity of the data, and the diversity of the generated data, thus providing a comprehensive evaluation of both the quality of data generation and the generative capacity of the model.

2. Evaluation of the generated data based on the classification capability of machine learning (ML) models: A subset of the original dataset is randomly selected as the test set, and the generated data is combined with a small amount of original data that are not included in the test set to form a training set for the machine learning model. A series of evaluation metrics are used to assess the model's performance on the test set, evaluate the impact of the generated data on the final model's training effectiveness, and reflect its effectiveness in real-world tasks.
3. T-SNE visualization was used to compare the generated data with the original data, evaluating the alignment between the generated data distribution and the true underlying distribution.

4.2.1. Similarity Evaluation Indicators

To evaluate distribution similarity, we use two metrics: Jensen-Shannon Divergence (JS) and Bhattacharyya Distance (BD). To assess morphological similarity, we use two metrics: Mean Squared Error (MSE) and Euclidean Distance (ED). For feature correlation and diversity, we use the Pairwise Similarity Difference (PSD) and the Diversity Score (DS), respectively.

1 Jensen-Shannon Divergence (JS): JS measures the similarity between the generated and original samples by calculating the Kullback-Leibler (KL) divergence between each distribution and their average distribution. JS is symmetric, bounded, and easy to interpret, with values closer to 0 indicating greater similarity between the two distributions. We sum the JS values across all columns and compute the average to obtain a compact and easily interpretable overall score.

$$\begin{aligned} KL(G,O) &= \sum_{i=1}^{n} \mathrm{softmax}(g_i) \log \frac{\mathrm{softmax}(g_i)}{\mathrm{softmax}(o_i)} \\ JS(G,O) &= \frac{1}{2}(KL(G,O) + KL(O,G)) \end{aligned} \tag{8}$$

2 Bhattacharyya Distance (BD): BD evaluates the similarity between the generated and original samples by calculating the degree of overlap be-

tween their distributions. BD is symmetric, with smaller values indicating a higher degree of overlap between the two distributions, thus reflecting greater distribution similarity. We sum the BD values across all columns and compute the average to obtain a compact and easily interpretable overall score.

$$BD(G,O) = -\ln\left(\sum_{i=1}^{n}\sqrt{\text{softmax}(g_i)\text{softmax}(o_i)}\right) \tag{9}$$

3 Mean Squared Error (MSE): MSE evaluates the similarity between the generated and original samples by computing the squared differences between them and averaging these values. A smaller MSE value indicates a smaller difference between the two samples, signifying higher similarity.

$$MSE(G,O) = \frac{1}{n}\sum_{i=1}^{n}(o_i - g_i)^2 \tag{10}$$

4 Euclidean Distance (ED): ED assesses the similarity between the generated and original samples by calculating the square root of the sum of squared differences between them in a multidimensional space. A smaller ED value suggests that the two samples are closer, implying greater similarity. We sum all ED values and compute their average to obtain a compact and easily interpretable overall score.

$$D_E(G,O) = \sqrt{\sum_{i=1}^{n}(o_i - g_i)^2} \tag{11}$$

5 Pairwise Similarity Difference (PSD): PSD evaluates the difference in feature correlations between the original and generated samples by first calculating the pairwise correlation matrices for both samples and then computing the L2 norm of the difference between these matrices. A smaller value indicates that the interaction differences between the features of the two samples are smaller, reflecting greater similarity in feature associations.

$$L2(a_o, a_g) = \sqrt{\sum_{i=1}^{n}(a_g - a_o)^2} \tag{12}$$

Where the matrices $\mathbf{a_o}$ and $\mathbf{a_g}$ represent the pairwise correlation matrices for the generated and original samples, respectively.

6 The Diversity Score (DS): DS assesses the diversity of the generated samples by calculating the average entropy (a measure of information content in information theory) of each feature column. A higher DS value indicates greater diversity in the generated data, with more complex and varied feature characteristics. We sum all DS values across the columns and compute their average to derive a compact and easily interpretable overall score.

$$DS(G) = -\sum_{i=1}^{n} \mathrm{softmax}(g_i) \ln \mathrm{softmax}(g_i) \tag{13}$$

4.2.2. Evaluation Indicators for Classification Ability of Machine Learning Models

We demonstrate the quality of the generated data through the classification capabilities of three widely used machine learning algorithms: Decision Tree (DT), Multilayer Perceptron (MLP), and Gradient Boosting Tree (XGBoost). To comprehensively evaluate model performance, we employ three primary quantitative metrics: Accuracy (ACC), F1-Score, and Area Under the Curve (AUC).

1 Accuracy (ACC): ACC evaluates the performance of the model by calculating the ratio of correctly classified samples to the total number of samples. A higher accuracy value indicates that more samples are correctly classified, suggesting better model performance.

$$ACC = \frac{TP + TN}{TP + FP + FN + TN} \tag{14}$$

Here, **TP** represents the samples where both the predicted and true values are normal; **FN** represents the samples where the predicted value is abnormal, but the true value is normal; **TN** represents the samples where both the predicted and true values are abnormal; and **FP** represents the samples where the predicted value is normal, but the true value is abnormal.

2 F1-Score: The F1-Score is the harmonic mean of Precision and Recall, used to assess the model's classification performance by balancing these two metrics. A higher F1-Score indicates a better balance between precision and recall, reflecting superior model performance.

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP+FP} \\ \text{Recall} &= \frac{TP}{TP+FN} \\ F1 &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \tag{15}$$

Here, **Predicted** represents the proportion of samples in which the predicted and true values are normal among all the samples predicted as normal, and **Recall** represents the proportion of samples where both the predicted and true values are normal among all the samples that are actually normal.

3 Area Under the Curve (AUC): AUC evaluates the discriminatory ability of the model by calculating the area under the ROC curve. The closer the AUC value is to 1, the better the model can distinguish between positive and negative classes, indicating stronger classification ability.

$$\begin{aligned} TPR &= \frac{TP}{TP+FN} \\ FPR &= \frac{FP}{FP+TN} \\ AUC &= \int_0^1 TPR(FPR)dFPR \end{aligned} \tag{16}$$

4.3. Result Analysis

4.3.1. Similarity Assessment

A subset of data from the original data set as a validation set to assess the similarity between the generated data and the real data. Specifically, five groups of data samples with comparable magnitudes were generated, and the similarity

Table 1: SWaT Similarity Evaluation Table

Method	Distributional Similarity		Morphological Similarity		Feature Correlation	Diversity
	JS	**BD**	**ED**	**MSE**	**PSD**	**Dive**
TimeGAN	0.00148	0.23953	1.81099	0.12775	45.89443	2.26511
TimeVAE	0.00178	0.28877	1.98298	0.22307	22.75481	**2.39109**
Diff-MTS	0.00400	0.64625	2.78790	0.16863	**9.43180**	2.21480
DiffWave	0.00634	1.02400	3.54311	0.46718	23.99984	2.21317
tabddpm	0.00613	0.99040	3.51883	0.45537	23.87467	2.22089
ETUA-DDPM	**0.00093**	**0.15042**	**1.40638**	**0.08593**	40.14236	2.24572

Notes: Best result in each column is highlighted in bold.

metrics between each group of samples and the original samples were calculated. Finally, the average of the similarity metrics across all groups was taken as the overall evaluation result, ensuring the stability and reliability of the evaluation process.

The experimental results on the SWaT dataset are shown in Table 1. First, in terms of distribution similarity, ETUA-DDPM achieves the lowest JS (0.00093) and BD (0.15042) values among all methods. Notably, the JS value is significantly better than those of TimeGAN (0.00148), Diff-MTS (0.00417), and DiffWave (0.00664), indicating that the distribution of the data generated by ETUA-DDPM most closely matches the real data. Similarly, in the morphology similarity evaluation, ETUA-DDPM yields lower ED (1.40638) and MSE (0.08593) values compared to other methods. In particular, ETUA-DDPM's MSE value is substantially lower than those of TimeVAE (0.22307) and tabddpm (0.45438), reflecting its precision in structural accuracy and its ability to preserve details in the generated data. Due to the feature correlation and diversity of the ETUA-DDPM decreased, with a PSD value of 40.14236 and a Dive value of 2.24572, but still maintained a level of diversity (the Dive value was slightly lower than TimeVAE (2.39109)). Overall, ETUA-DDPM demonstrates superior comprehensive performance in terms of distribution and morphology similarity, feature correlation, and data diversity, reflecting its stability and efficiency in generation tasks.

The experimental results on the WADI dataset are shown in Table 2. The

Table 2: WADI Similarity Evaluation Table

Method	Distributional Similarity		Morphological Similarity		Feature Correlation	Diversity
	JS	**BD**	**ED**	**MSE**	**PSD**	**Dive**
TimeGAN	0.00157	0.03170	42.35706	**0.14418**	118.54401	1.83935
TimeVAE	0.00202	0.04053	47.53767	0.21683	33.14430	2.65602
Diff-MTS	0.00319	0.06358	60.55264	0.42327	27.87778	3.17178
DiffWave	0.00314	0.06268	59.95979	0.42472	27.62304	**3.23098**
tabddpm	0.00320	0.06374	60.43619	0.42668	**27.46864**	3.16298
ETUA-DDPM	**0.00156**	**0.03140**	**41.86870**	0.28291	28.55125	1.84562

Notes: Best result in each column is highlighted in bold.

ETUA-DDPM method performs similarly or better across multiple metrics compared to other methods. Specifically, in terms of distribution similarity metrics (JS and BD), ETUA-DDPM achieves the lowest values of 0.00156 and 0.03140, respectively, which are significantly better than Diff-MTS (0.00319 and 0.06358) and TimeVAE (0.00202 and 0.04053), demonstrating higher distribution accuracy for the generated data. Regarding morphology similarity metrics (ED and MSE), ETUA-DDPM also performs excellently, with an MSE of 0.28291, slightly higher than TimeGAN (0.14418), and an ED of 41.86870(the smallest value), which proves that ETUA-DDPM retains a high level of precision in the structural form of the generated data. Moreover, ETUA-DDPM's feature correlation (PSD) is 28.55125, which is close to tabddpm's (27.46864), indicating that the method is effective in preserving the correlations between features of the real data. Similarly, due to the improvement in similarity, ETUA-DDPM's Dive is 1.84562, indicating a slight reduction in diversity. Overall, ETUA-DDPM not only ensures that the generated data is similar to the real data in terms of distribution and morphology but also effectively controls the correlations between features while maintaining a reasonable level of diversity, demonstrating strong performance.

4.3.2. Evaluation of Classification Ability Based on Machine Learning Models

We conducted five independent experiments and calculated the average result for each evaluation metric, which was used as the final performance evalua-

Table 3: Comparison of the effects of different generation methods on different classifiers

Method	**SWaT**			**WADI**		
	ACC(%)	**F1-score(%)**	**AUC(%)**	**ACC(%)**	**F1-score(%)**	**AUC(%)**
DT						
TimeGAN	94.1327	94.1125	94.1327	**92.5721**	**92.5435**	**92.5737**
TimeVAE	92.9683	92.7251	92.9683	82.7654	79.3734	82.7646
Diff-MTS	63.9291	58.5339	63.9291	92.4617	92.4302	92.4634
DiffWave	50.3167	34.0334	50.3167	91.4881	91.4340	91.4901
tabddpm	50.3167	34.0334	50.3167	92.1807	92.1326	92.1826
ETUA-DDPM	**94.3258**	**94.3074**	**94.3258**	64.2409	57.2803	64.2335
MLP						
TimeGAN	**99.9246**	**99.9246**	**99.9653**	83.0113	82.4503	**99.9968**
TimeVAE	**99.9246**	**99.9246**	99.9649	82.3287	81.6517	99.9029
Diff-MTS	**99.9246**	**99.9246**	99.9620	91.6688	91.6048	99.9555
DiffWave	94.1735	94.1536	99.9621	88.7378	88.5812	99.9943
tabddpm	94.1569	94.1368	99.9577	90.7102	90.6190	99.9959
ETUA-DDPM	**99.9246**	**99.9246**	99.9648	**94.6248**	**94.5982**	99.9773
XGBoost						
TimeGAN	94.1327	94.1125	96.6199	92.1757	92.1277	99.8326
TimeVAE	**95.1448**	**95.1326**	99.7616	71.0866	63.8338	99.6065
Diff-MTS	94.1327	94.1125	94.9084	92.3112	92.2654	99.8866
DiffWave	94.3439	94.3257	**99.9345**	91.5583	91.4986	99.9049
tabddpm	94.3439	94.3257	99.9311	92.5872	92.5476	**99.9063**
ETUA-DDPM	94.3831	94.3652	98.4706	**93.8269**	**93.8107**	97.9709

Notes: Best result in each column is highlighted in bold.

tion. Table 3 presents the performance evaluation results of different generative methods in multiple machine learning models.

First, on the SWaT dataset, the data generated by ETUA-DDPM exhibited outstanding performance on the DT model, achieving the best values in ACC (94.3258%), F1 score (94.3258%), and AUC (94.3074%), slightly surpassing TimeGAN (94.1327%, 94.1327%, and 94.1125%), and significantly outperforming TimeVAE, Diff-MTS, DiffWave, and tabddpm. On the MLP and XGBoost models, ETUA-DDPM performed comparable to TimeGAN and TimeVAE, and outperformed Diff-MTS, DiffWave, and tabddpm. Next, on the WADI

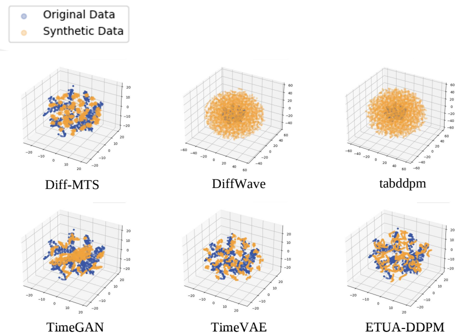

Figure 4: t-SNE Visualization on SWaT Dataset.

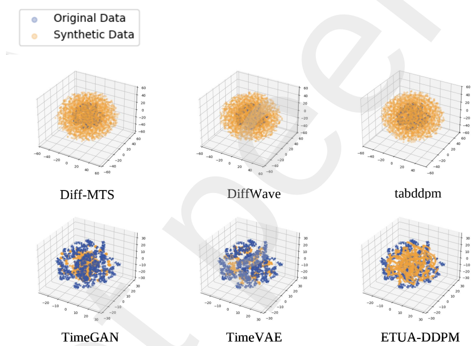

Figure 5: t-SNE Visualization on WADI Dataset.

dataset, ETUA-DDPM achieved the best values in ACC (94.6248%) and AUC (94.5982%) on the MLP model, showing an average improvement of 9%. Similarly, on the XGBoost model, it achieved the best results in ACC (93.8269%) and AUC (93.8107%), significantly outperforming other generative methods.

4.3.3. T-SNE Visualization

We selected 3000 corresponding samples of real and generated abnormal data from both the SWaT and WADI datasets and visualized them using the T-SNE technique, as shown in Fig. 4 and Fig. 5. In the T-SNE visualization for SWaT, it can be observed that the IMATD generated by Diff-MTS, TimeVAE, and

ETUA-DDPM more closely approximate the true distribution. In the T-SNE visualization for WADI, the IMATD generated by ETUA-DDPM demonstrates a superior fit to the true distribution.

Overall, ETUA-DDPM is capable of generating abnormal samples that are highly consistent with the global distribution of real IMATD. This achieves effective augmentation of scarce ICS abnormal data, thereby enhancing the performance and applicability of downstream detection models.

4.4. Ablation Experiment

To further validate the effectiveness of the ETUA-DDPM method, we designed an ablation experiment progressively retaining and removing different improved modules to evaluate their contributions to model performance. Specifically, the experiment includes the following two setups:

1. **Ori-DDPM**: The original diffusion model;
2. **ETU-DDPM**: Retaining only the modifications made to the encoder and decoder parts, without introducing the AASC mechanism;
3. **ETUh-DDPM**: Modifications are made to the encoder part, along with the introduction of the AASC mechanism.

During the experiment, we generated five sets of comparable data samples from the SWaT dataset and evaluated the performance of each model by calculating similarity metrics. To ensure the reliability of the results, we averaged the evaluation results for each sample set and ultimately compared them with the performance of ETUA-DDPM. The comparison results, as shown in Table 4, demonstrate the performance differences among the various models in the data generation task.t Based on the data presented in the table, it is observed that ETUA-DDPM significantly outperforms the other three methods in terms of distribution similarity and morphological similarity metrics. Specifically, ETUA-DDPM achieves lower values in JS (0.00093), BD (0.15042), ED (1.40638), and MSE (0.08593), indicating that the distribution of its generated samples has the smallest difference compared to the real data. Providing strong

Table 4: Comparison of similarity in ablation experiments

Method	Distributional Similarity		Morphological Similarity		Feature Correlation	Diversity
	JS	BD	ED	MSE	PSD	Dive
Ori-DDPM	0.00111	0.17997	1.60161	0.10333	41.98027	2.27286
ETU-DDPM	0.00677	1.09455	4.55031	0.21817	49.85274	**3.45200**
ETUh-DDPM	0.00146	0.23669	1.84766	0.11737	44.45062	2.62000
ETUA-DDPM	**0.00093**	**0.15042**	**1.40638**	**0.08593**	**40.14236**	2.24571

Notes: Best result in each column is highlighted in bold.

Table 5: Comparison of classifier performance in ablation experiments

Model	Method	ACC(%)	F1-score(%)	AUC(%)
DT	Ori-DDPM	58.825	45.489	58.825
	ETU-DDPM	76.480	76.480	69.801
	ETUh-DDPM	88.097	88.097	86.863
	ETUA-DDPM	**94.326**	**94.326**	**94.307**

Notes: Best result in each column is highlighted in bold.

evidence of the higher quality and effectiveness of the generated data. Moreover, ETUA-DDPM achieves the lowest value in PSD (40.14236), suggesting that the correlations between the features of the generated data are closer to those of the real data. Furthermore, although the diversity (Dive) value of ETUA-DDPM is 2.24571, slightly lower than that of ETU-DDPM (3.45200), its overall performance remains superior. This indicates that ETUA-DDPM is capable of providing more stable and task-appropriate generated samples while maintaining relatively low similarity.

We also performed comparative experiments to evaluate the classification abilities of machine learning models. This study presents the comparative results based on the DT model, where Ori-DDPM and ETU-DDPM exhibited a phenomenon in which the prediction results focused on a single class, leading to lower values. Based on the results of the ablation experiments in Table 5, the ETUA-DDPM method demonstrates a significant performance improvement compared to the other three methods, surpassing the other three methods in all three metrics. In the decision tree (DT) model, ETUA-DDPM outperforms ETU-DDPM (76.480%, 76.480%, 69.801%) and ETUh-DDPM (88.097%, 88.097%, 86.863%) by a substantial margin in ACC (94.326%), F1-score (94.326%), and AUC (94.307%), indicating that ETUA-DDPM enhances the classifier's classification capability.

It is evident that each improved module within ETUA-DDPM plays a crucial role in enhancing the model's performance. Specifically, the ETU network strengthens the model's ability to capture complex temporal relationships by enhancing long temporal dependency modeling, while the AASC effectively improves the capability to preserve detailed features. These enhancements synergistically optimize the generation process, further improving the quality of the generated abnormal time series data. Therefore, the cooperative function of the various components of ETUA-DDPM significantly boosts the overall model's performance.

5. CONCLUSION

To address the core challenge of extreme scarcity of real anomalous samples in Industrial Control System (ICS) anomaly detection, this paper proposes ETUA-DDPM, a Denoising Diffusion Probabilistic Model (DDPM) that integrates the state space model Mamba with an adaptive attention mechanism. By enhancing the capability to model long temporal dependencies and preserve fine-grained features in IMATD, the model achieves the generation of the anomalous samples that maintain high consistency with the global distribution of real anomalous data. This effectively addresses the bottleneck issue of anomalous data scarcity in ICS scenarios. Validation through two categories of experiments and visualization methods demonstrates that ETUA-DDPM can successfully augment scarce anomalous samples in ICS, thereby improving the performance of ICS anomaly detection models and providing reliable data support for enhancing the operational effectiveness of ICS security systems.

While the method proposed in this study effectively augments the scarce abnormal samples in ICS, it is worth acknowledging that the current work still has certain limitations. For instance, the capacity for modeling complex inter-variable correlations needs to be deepened, there is a lack of explicit causal inference mechanisms, and the flexibility of conditional-controlled generation is insufficient. Future research will focus on exploring these areas in depth to further enhance the method's performance and applicability.

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